ABDURAHIMAN THOMMANGADAN

Node.js Backend Engineer | NestJS, Microservices, Kafka, gRPC, PostgreSQL, Generative AI

Bangalore, India | +91 79073 82644 | abdurahmant7@gmail.com | **LinkedIn**

PROFESSIONAL SUMMARY

Node.js Backend Engineer with 7+ years building scalable, secure microservices and data-intensive, event-driven systems on Node.js (Express.js, NestJS), PostgreSQL, Redis, Kafka and gRPC, including Samsung.com global eCommerce (Samsung SDS), enterprise SaaS (Infosys) and VData, a distributed platform with near-instant search across 30 million records. Technical Lead at Voizzit since 2024, delivering CADBEATER end to end with LLMs, RAG and LangGraph agents. Strong in Clean (Onion) Architecture, scaling, JWT/OAuth and RBAC security, code reviews, CI/CD, Docker and Kubernetes.

TECHNICAL SKILLS

Backend & Microservices: Node.js, Express.js, NestJS, TypeScript, JavaScript (ES6+), Microservices, RESTful API Design, gRPC, WebSocket, Server-Sent Events (SSE), ORM (Sequelize, TypeORM), Python, FastAPI

Event-Driven & Messaging: Apache Kafka (event processing), RabbitMQ, Redis Streams, Event-Driven Architecture, Asynchronous Workers & Queues

Databases & Performance: PostgreSQL, MongoDB, MySQL, Redis (caching), Qdrant, High-Volume Data Processing, Horizontal & Vertical Scaling

Observability: Logging & Monitoring, ELK Stack (Elasticsearch), End-to-End Traceability, Langfuse (LLM tracing), Cost & Latency Monitoring

Security: JWT & OAuth 2.0 Authentication, Role-Based Access Control (RBAC), OWASP Top 10, Secure API Design, Rate Limiting, Audit Logging, Encryption in Transit & at Rest

Architecture & Quality: Clean (Onion) Architecture, SOLID & Design Patterns, System Design, Distributed Systems, High Availability, API Gateway, TDD, BDD, Code Reviews, Jest, Supertest, Playwright

Cloud & DevOps: AWS (EC2, S3, SQS), Docker, Kubernetes, CI/CD, GitHub Actions, ELK Stack

Generative AI & LLMs: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangGraph, LangChain, AI Agents, Model Context Protocol (MCP), OpenAI, Anthropic (Claude), OpenRouter, vLLM, Prompt Engineering, Agentic Coding (Claude Code)

Frontend & Mobile: React.js, Angular, Redux Toolkit, TanStack Query, React Native, Flutter

PROFESSIONAL EXPERIENCE

Voizzit Information Technology LLC, Dubai, UAE — Sep 2024 – Present

Technical Lead — Full Stack & Generative AI

Flagship project: CADBEATER

- Led CADBEATER from requirements through production with a modular, horizontally scalable design across the multi-agent AI backend, FastAPI services, MongoDB data layer and React frontend. Its estimator generates cost estimates, bills of quantities (BOQ), bills of materials (BOM) and CAD drawings from requirements, joinery details or floor plans, reducing manual estimating effort from four hours to ten minutes (about 96% less time).
- Built the LLM layer: LangGraph agents (ReAct, Reflexion, Plan-and-Execute), RAG over Qdrant, Model Context Protocol (MCP) tool integration and short- and long-term agent memory, with LLMs from OpenAI, Anthropic (Claude) and OpenRouter plus a self-hosted vision-language model on vLLM, task-based model routing, caching and structured outputs.
- Engineered for scale and reliability: event-driven FastAPI services on Redis Streams, GPU-isolated inference workers, and real-time SSE streaming to the React CAD frontend (Redux Toolkit, TanStack Query, Three.js).
- Secured the platform with role-based access control (RBAC), role- and team-scoped data access, audit logging and human-in-the-loop approval gates.
- Owned quality, code reviews and observability: Langfuse tracing, prompt versioning, evaluation pipelines, cost and latency monitoring and an AI-driven SDLC (agentic coding with Claude Code, custom agent skills); authored the architecture reference adopted across engineering and product.

Senior Backend Engineer — SaaS Platforms

- Architected VData, a distributed voter-data platform, on Node.js and NestJS with Clean (Onion) Architecture, separating domain, application and infrastructure layers, with TypeORM over PostgreSQL.

- Scaled VData horizontally and vertically on AWS to enable fast, near-instant search across 30 million voter records; supported the platform with ELK analytics, CI/CD and automated API test suites.
- Built TechBuddy LMS (admin CMS, student web, React Native TV apps) on Node.js, NestJS and PostgreSQL, with AI-generated quizzes, offline content export, secure authentication and rate limiting.

Project: AutoFleet — Car Rental Booking & ERP Platform

- Built a car rental booking and ERP platform covering the end-to-end booking flow, fleet management, reservations, CRM and HRM modules.
- Delivered a cross-platform Flutter mobile app for Android, iOS and web, and handled release management and ongoing maintenance on the Google Play Store and Apple App Store.
- Built the Node.js and React.js web platform on MongoDB and AWS, with Playwright-based automation for third-party integrations.

Tech stack: Node.js, React.js, Flutter (Dart), MongoDB, AWS, Playwright

Voizzit tech stack: Node.js, NestJS, TypeScript, PostgreSQL, MongoDB, Redis, Docker, Kubernetes, CI/CD, AWS, ELK Stack, LLMs, RAG, LangGraph, FastAPI, React.js, Angular

Samsung SDS India, Chennai, India — Feb 2024 – Sep 2024

Backend Engineer — Samsung.com Global eCommerce

- Built and operated independently deployable Node.js microservices for Samsung.com global eCommerce, supporting autoscaling during high-traffic sales events; used Kafka and RabbitMQ for event processing and PostgreSQL for transactional data.
- Built secure logistics and user-interaction APIs with Elasticsearch logging and monitoring for end-to-end traceability.

Infosys Limited, Bangalore, India — Jun 2022 – Feb 2024

Associate Consultant — Enterprise SaaS

- Delivered full-stack features and high-availability microservices for Solargraf, a solar-design SaaS with 3D design and financial modules (Node.js, Express.js, React.js, Sequelize, PostgreSQL).
- Built and tested microservices behind an API Gateway with gRPC, and took part in architecture reviews with cross-functional product teams.

Kimball Electronics India Pvt. Ltd., Trivandrum, India — Nov 2020 – Jun 2022

Software Engineer — Industrial IoT

- Built the Alerting, Visualization and Data Acquisition modules of EMTAB, an Industry 4.0 IoT predictive-maintenance platform (Node.js, React.js, Express.js, Sequelize, MongoDB, MySQL).
- Integrated machine-learning anomaly detection into real-time shop-floor sensor pipelines, and wrote Playwright test automation.

Nucore Software Solutions, Calicut, India — Apr 2019 – Nov 2020

Backend Developer — Travel Accounting SaaS

- Designed and built the ERP accounting engine, microservice APIs and business logic for TRAACS, a back-office SaaS for travel agencies (Node.js, Express.js, Angular, PostgreSQL, MongoDB).

EDUCATION

B.Sc. Computer Science, Farook College, Kozhikode (University of Calicut) — 2016 – 2019

CERTIFICATIONS

- Machine Learning Specialization — DeepLearning.AI (Andrew Ng)
- Deep Learning — Codebasics